

Configure **Load Balancing**

Azure Load Balancer, Application Gateway, Traffic Manager, and Azure Front Door — which one to use, how each one works, and the key configuration components the exam tests.

Four load balancing services — pick the right one

Azure has four distinct load balancing services. The exam tests whether you can match the right service to the scenario — Layer 4 vs Layer 7, internal vs external, regional vs global.

LAYER 4 • REGIONAL

Azure Load Balancer

- TCP/UDP traffic distribution
- Internal (private) or External (public)
- Health probes: TCP, HTTP, HTTPS
- No SSL termination or URL routing
- Best for: VMs, VMSS, non-HTTP workloads

LAYER 7 • REGIONAL

Application Gateway

- HTTP/HTTPS/WebSocket routing
- URL path-based and host-based routing
- SSL/TLS termination and re-encryption
- Web Application Firewall (WAF) add-on
- Best for: web apps, API gateways, WAF

DNS-BASED • GLOBAL

Traffic Manager

- DNS-level routing to global endpoints
- Routing: performance, priority, weighted, geographic
- Works with any internet-reachable endpoint
- No data plane — only DNS responses
- Best for: multi-region failover, geographic routing

LAYER 7 • GLOBAL

Azure Front Door

- Global HTTP/HTTPS load balancing via Anycast
- CDN, WAF, SSL offload at the edge
- URL path routing, caching, session affinity
- Sub-second failover vs Traffic Manager's DNS TTL
- Best for: global web apps, low latency at scale

Azure Load Balancer anatomy

FRONTEND IP

Public or Internal IP

The address clients connect to

BACKEND POOL

VMs / VMSS / NICs

The targets that receive traffic

HEALTH PROBE

TCP / HTTP / HTTPS

Marks backends healthy or unhealthy

↓ traffic flows from Frontend → Load Balancing Rule → healthy Backend Pool members

LOAD BALANCING RULE

Protocol + Frontend Port →
Backend Port

Maps incoming traffic to the
backend pool

INBOUND NAT RULE

Frontend Port → specific VM
+ port

Direct access to individual
backend VMs

SESSION PERSISTENCE

None / Client IP / Client
IP+Protocol

Sticky sessions — same client
→ same backend

Eight terms you need cold

+ Health Probe

Periodically tests each backend to determine if it's healthy enough to receive traffic. Failed probes remove the instance from rotation until it recovers.

Tip: If all backends fail the probe, all traffic is sent to all backends (fail-open). Design probes to test real app health, not just TCP connectivity.

Backend Pool

The set of VMs, NIC IP configs, or VMSS instances that receive load-balanced traffic. Standard LB supports VMs in any availability zone in the region.

Tip: Standard SKU backends must also use Standard SKU Public IPs — mixing SKUs is not supported.

> Session Persistence

Ensures all requests from the same client go to the same backend VM. Options: None (default, 5-tuple hash), Client IP (2-tuple), Client IP + Protocol (3-tuple).

Tip: Session persistence is not the same as sticky sessions at the application layer — use Application Gateway cookies for true app-layer affinity.

Listener (App Gateway)

A configuration on Application Gateway that monitors a specific IP, port, protocol, and hostname for incoming requests — the entry point for routing rules.

Tip: Multi-site listeners let one Application Gateway route requests for multiple hostnames (e.g. app1.contoso.com vs app2.contoso.com) to different backend pools.

WAF (Web Application Firewall)

An optional add-on to Application Gateway and Front Door that inspects HTTP traffic for common web exploits (OWASP rule sets: SQLi, XSS, etc.) — runs in Detection or Prevention mode.

Tip: Always start WAF in Detection mode to baseline false positives before switching to Prevention (blocking) mode.

Traffic Manager Profile

A DNS-based load balancer with routing methods: Performance (lowest latency), Priority (primary/failover), Weighted (percentage split), Geographic (by client region), Multivalued, Subnet.

Tip: Traffic Manager only returns a DNS answer — it doesn't proxy traffic. Failover time = DNS TTL (minimum 0 seconds, default 300).

→ Outbound Rules (Standard LB)

Explicit SNAT (Source Network Address Translation) rules for outbound connections from VMs in the backend pool — required when Standard LB has no public load balancing rule for a VM.

Tip: Basic LB provides automatic SNAT. Standard LB requires explicit outbound rules or a NAT Gateway — VMs with no outbound rule and no public IP lose internet access.

🔗 Floating IP (Direct Server Return)

Sends packets to the backend with the frontend IP as the destination — used for SQL Always On Availability Groups and other scenarios requiring multiple VIPs on one NIC.

Tip: When Floating IP is enabled, the OS on the backend VM must be configured to respond to the frontend IP (loopback adapter on Windows).

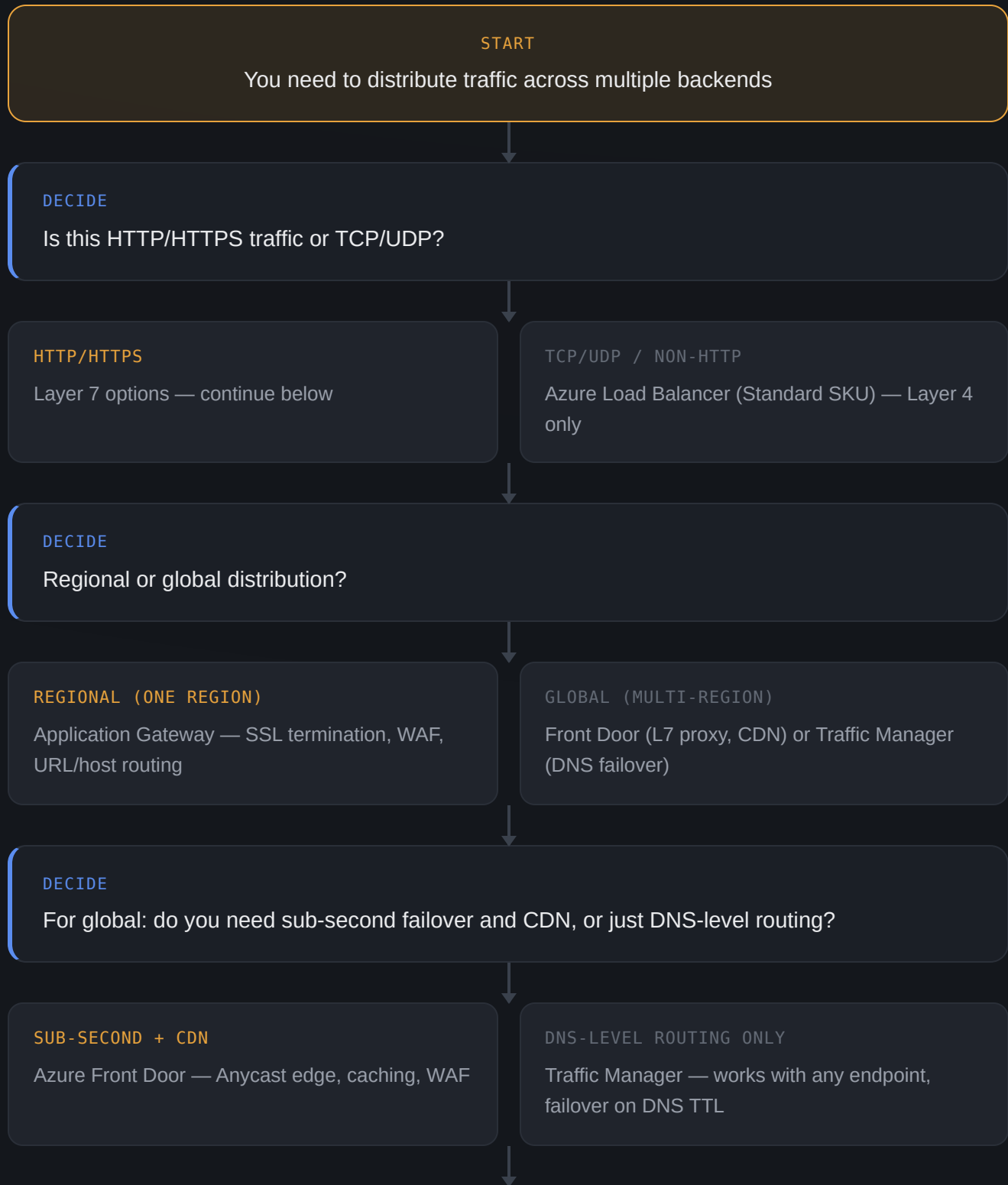
SERVICE	LAYER	SCOPE	SSL TERMINATION	WAF	BEST FOR
Load Balancer	L4 (TCP/UDP)	Regional	No	No	VMs, VMSS, internal services
Application Gateway	L7 (HTTP/S)	Regional	Yes	Yes	Web apps, URL routing, WAF
Traffic Manager	DNS	Global	No	No	Multi-region failover, geo routing
Front Door	L7 (HTTP/S)	Global	Yes	Yes	Global web apps, CDN, low latency

EXAM GOTCHA — STANDARD LB LOSES AUTOMATIC OUTBOUND

Basic Load Balancer provides automatic SNAT for backend VMs to reach the internet. Standard Load Balancer **does not** — VMs in a Standard LB backend pool with no public IP and no explicit outbound rule **lose all outbound internet access**. Fix it with an outbound rule, a NAT Gateway on the subnet, or a public IP on each VM.

Choosing and configuring the right load balancer

Every load balancing scenario comes down to three questions: what layer, what scope, and what features.



DECIDE

Should the load balancer be internet-facing or internal only?

INTERNET-FACING

Public frontend IP — Standard SKU required

INTERNAL ONLY

Internal Load Balancer — private frontend IP in subnet

CONFIGURE

Frontend IP → Backend Pool → Health Probe → Load Balancing Rule → (Outbound Rules for Standard LB)

DONE

Traffic distributed across healthy backends with the right service, tier, and rules

Tools for load balancer configuration

>_ Azure CLI — Load Balancer

BEST FOR: scripted LB setup from scratch

Create a Standard Load Balancer with frontend IP, backend pool, probe, and rule in one script — fully automatable for pipeline deployments.

```
az network lb create \
  -g rg-prod -n lb-web \
  --sku Standard \
  --public-ip-address pip-lb-web \
  --frontend-ip-name fe-web \
  --backend-pool-name be-web

az network lb probe create \
  -g rg-prod --lb-name lb-web \
  -n probe-http --protocol Http \
  --port 80 --path /health

az network lb rule create \
  -g rg-prod --lb-name lb-web \
  -n rule-http --protocol Tcp \
  --frontend-port 80 --backend-port
80 \
  --frontend-ip-name fe-web \
  --backend-pool-name be-web \
  --probe-name probe-http
```

>_ Azure CLI — Application Gateway

BEST FOR: scripted App Gateway creation

Create an Application Gateway with WAF in one command — specify backend pools, HTTP settings, and routing rules.

```
az network application-gateway
create \
  -g rg-prod -n agw-web \
  --sku WAF_v2 --capacity 2 \
  --vnet-name vnet-prod \
  --subnet agw-subnet \
  --public-ip-address pip-agw \
  --http-settings-port 80 \
  --http-settings-protocol Http \
  --frontend-port 443 \
  --cert-file cert.pfx \
  --cert-password "Password123"

az network application-gateway waf-
config set \
  -g rg-prod --gateway-name agw-web
\
  --enabled true --firewall-mode
Prevention \
  --rule-set-version 3.2
```

> Azure CLI — Traffic Manager

BEST FOR: multi-region DNS failover

Create a Traffic Manager profile and add regional endpoints for priority-based or performance-based failover in minutes.

```
az network traffic-manager profile
create \
  -g rg-prod -n tm-web \
  --routing-method Priority \
  --dns-config-relative-name myapp-
global \
  --ttl 30

az network traffic-manager endpoint
create \
  -g rg-prod --profile-name tm-web
\
  -n ep-eastus --type
azureEndpoints \
  --target-resource-id $(az webapp
show \
  -g rg-prod -n myapp-eastus \
  --query id -o tsv) \
  --endpoint-status Enabled \
  --priority 1
```

<> Bicep — Standard Load Balancer

BEST FOR: IaC — LB as code

Deploy the full load balancer stack alongside backend VMs in one template — probes, rules, and outbound rules all version-controlled.

```
resource lb
'Microsoft.Network/loadBalancers@20
23-04-01' = {
  name: 'lb-web'
  location: location
  sku: { name: 'Standard' }
  properties: {
    frontendIPConfigurations: [{
      name: 'fe-web'
      properties: {
        publicIPAddress: { id: pip.id } }
    }]
    backendAddressPools: [{ name:
'be-web' }]
    probes: [{
      name: 'probe-http'
      properties: { protocol:
'Http', port: 80, requestPath:
'/health' }
    }]
    loadBalancingRules: [{
      name: 'rule-http'
      properties: {
        frontendIPConfiguration: {
          id: resourceId('...', 'fe-web') }
        backendAddressPool: { id:
resourceId('...', 'be-web') }
        probe: { id:
resourceId('...', 'probe-http') }
        protocol: 'Tcp'
        frontendPort: 80
        backendPort: 80
      }
    }]
  }
}
```


SERVICE	CLI NAMESPACE	KEY CREATE COMMAND
Load Balancer	<code>az network lb</code>	<code>az network lb create</code>
Application Gateway	<code>az network application-gateway</code>	<code>az network application-gateway create</code>
Traffic Manager	<code>az network traffic-manager</code>	<code>az network traffic-manager profile create</code>
Front Door	<code>az afd</code>	<code>az afd profile create</code>

Principles worth internalizing

WHEN TO COMBINE SERVICES

Traffic Manager and Front Door can sit in front of Application Gateway or Load Balancer — e.g. Traffic Manager routes globally to the nearest region's Application Gateway, which then handles WAF and URL routing to VMs. **Front Door + Application Gateway** is a common pattern: Front Door handles global Anycast + CDN, App Gateway handles regional WAF and routing. Never put Traffic Manager in front of Front Door — they serve the same global routing purpose.

EXAM GOTCHA — ALL BACKENDS FAIL THE HEALTH PROBE

If **all** backend instances fail the health probe, Azure Load Balancer uses a **fail-open** behavior and sends traffic to all backends anyway — it's better than dropping all traffic. This is unique to Azure LB. Application Gateway fails closed — if all backends are unhealthy, it returns a 502 Bad Gateway.

Six mistakes that show up on the exam

Using Basic LB when Standard features are needed

Basic LB doesn't support Availability Zones, outbound rules, or mixing with Standard SKU Public IPs. Use Standard for all new production deployments.

Forgetting Standard LB requires explicit outbound rules

VMs behind a Standard LB with no public IP and no outbound rule lose internet access. Add an outbound rule or deploy a NAT Gateway on the subnet.

Using Traffic Manager when sub-second failover is needed

Traffic Manager failover is bound by DNS TTL (minimum ~0s, but caches vary). For near-instant global failover, use Azure Front Door which uses Anycast and health probes at the edge.

Enabling WAF in Prevention mode immediately

Always start in Detection mode, review logs for false positives, then switch to Prevention. Going straight to Prevention can block legitimate traffic and is very hard to debug in production.

Expecting Application Gateway to work without a dedicated subnet

Application Gateway requires its own dedicated subnet — no other resources (VMs, other services) should share the App Gateway subnet.

Mixing Basic and Standard SKU resources

Standard SKU Load Balancers require Standard SKU Public IPs — Basic public IPs cannot be assigned to a Standard LB frontend. All resources in a tier must match SKU.